

Correcting Fund Performance Rankings Under Regime Difficulty with Item Response Theory

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Abstract

We apply the Evaluation Failure Scaling Law (EFSL) framework to global fund performance assessment, showing that simple averaging of binary monthly outperformance can produce substantial model-implied ranking distortions under regime-dependent period difficulty. Using a 2-parameter logistic Item Response Theory (IRT) model on 50 exchange-traded funds across 217 monthly periods (April 2008 to April 2026), we find that IRT corrections shift fund rankings by up to ± 30 positions (Spearman $\rho = 0.573$). Crisis periods exhibit lower IRT difficulty than normal periods (month-level Mann–Whitney $p = 0.001$; contiguous-block permutation $p = 0.055$), consistent with the interpretation that market downturns are easier periods for relative outperformance. The Linear Logistic Test Model provides an exploratory decomposition of period difficulty, though cross-validated performance is weak ($R^2_{CV} < 0$). Principal Component Regression indicates a dominant common return component (PC1 = 59.2%). A sparsity stress test with a 1PL Rasch refit compares rankings under artificial masking against the full-data 2PL ranking, which is itself an IRT estimate rather than an external ground truth. The canonical run recorded the 1PL refit ahead of simple averaging at 7 of 10 sparsity levels (0–90%, including the unmasked level), but its per-level values were not committed; an offline refit gives 6 of 10, with IRT advantages of +0.003 to +0.014 in Spearman ρ and losses at 50%, 70%, 80%, and 90% sparsity (down to -0.041 at 90%). This provides an additional cross-domain application of the EFSL framework.

Keywords: Item Response Theory, Fund Evaluation, Market Regimes, EFSL

1 Introduction

Fund performance evaluation relies on simple average returns, Sharpe ratios, and Jensen’s alpha [Sharpe, 1966, Jensen, 1968]. These methods assume all evaluation periods contribute equally—an assumption that fails when funds operate across market regimes of varying difficulty and not all funds exist in all periods.

The Evaluation Failure Scaling Law [Kang, 2026] demonstrates that simple averaging produces unreliable rankings under high sparsity and heterogeneous item difficulty. We extend this framework to finance with the structural mapping: subjects (θ) = 50 global ETFs; items (b) = 217 monthly periods; binary response = did the fund beat ACWI in that month? Because most ETFs in our universe are passive index trackers rather than actively managed funds, the latent “fund ability” parameter θ should be interpreted as a benchmark-relative outperformance propensity driven by asset-class exposure, not as a measure of active management skill.

We apply three tools: (1) IRT (2PL) to jointly estimate fund ability and period difficulty with post-optimization centering; (2) LLTM to explore whether macro features explain period difficulty; (3) PCR to extract latent market factors.

2 Data and Methods

2.1 Data

Daily adjusted closing prices for 50 U.S.-listed ETFs and the MSCI ACWI ETF benchmark are collected via Yahoo Finance. The sample begins April 2008, after the ACWI ETF’s launch date of March 26, 2008, to ensure a consistent benchmark throughout. The sample ends April 2026, yielding 217 monthly periods.

Table 1: ETF universe by region.

Region/Class	N	Examples
US Broad & Growth	7	SPY, QQQ, IWM, DIA, ARKK, VTI, VOO
US Sectors	11	XLK, XLF, XLE, XLV, XLRE, SMH, XLU, XLI, XLP, XLB, XLC
Europe	9	EWG, EWU, EWQ, EWJ, EWP, EWL, EWN, EWD, EWK
Asia-Pacific	7	EWJ, EWY, EWA, EWT, EWS, EWH, MCHI
Emerging Markets	8	EEM, EWZ, IND, EWW, TUR, EZA, ECH, EPOL
Commodities/Crypto	3	GLD, SLV, IBIT
Fixed Income/REITs	5	TLT, HYG, LQD, TIP, VNQ
Total	50	

The binary response matrix has 10,233 observed entries out of 10,850 possible (94.3% coverage, 5.7% natural sparsity). Missing entries arise when funds did not yet exist (e.g., IBIT launched January 2024).

Data availability. Raw price data is sourced from Yahoo Finance via the `yfinance` Python library, which is intended for personal and research use. We provide exact tickers and date ranges in a manifest file; users must download data themselves. This study is for research purposes only and does not constitute investment advice. Yahoo Finance data are informational and may be revised; users should verify data independently.

2.2 IRT (2PL) Model

We fit a 2-parameter logistic IRT model: $P(X_{ij} = 1 \mid \theta_i, a_j, b_j) = [1 + \exp(-a_j(\theta_i - b_j))]^{-1}$, where θ_i is fund ability, b_j is period difficulty, and $a_j > 0$ is discrimination. Parameters are estimated by L-BFGS-B with analytic gradients.

Identification. The 2PL likelihood is invariant to a joint location shift of θ and b . We resolve this indeterminacy by post-optimization centering: $\theta_i \leftarrow \theta_i - \bar{\theta}$ and $b_j \leftarrow b_j - \bar{b}$. Scale is controlled operationally by bounding log-discrimination parameters ($|\log a_j| \leq 5$). All rank-based fund comparisons and period-difficulty comparisons are invariant to this additive centering; absolute parameter magnitudes should not be over-interpreted.

Assumptions. IRT assumes unidimensionality, local independence, and monotonicity. In finance, unidimensionality is approximate: a single “ability to outperform” dimension captures most variance but ignores multi-factor style exposures. Local independence is violated by temporal autocorrelation between adjacent months within the same market regime. We partially address this via the contiguous-block permutation test (Section 3.2). Formal goodness-of-fit diagnostics (e.g., Q_3 statistics for residual dependence) are deferred to future work.

2.3 LLTM, PCR, and Crisis Validation

The LLTM decomposes $b_j = \sum_f q_{jf} \eta_f + \epsilon_j$ using 5 features (VIX, dispersion, ACWI return, credit spread, crisis flag) via Ridge regression with 10-fold CV.

PCA is applied to the standardized return matrix; the first k components predict IRT difficulty via OLS.

Seven known crises serve as external validation. Because adjacent crisis months are autocorrelated, we supplement the month-level Mann–Whitney test with a **contiguous-block permutation test**: for each of the 7 crisis episodes, we compute the mean b across its months, then repeatedly sample contiguous blocks of the same length from the full time series (10,000 permutations), comparing the observed crisis-episode mean against the null distribution.

3 Results

3.1 IRT Ranking Corrections

The 2PL model converges in 740 iterations. Spearman $\rho = 0.573$ ($p < 10^{-4}$), Kendall $\tau = 0.402$ ($p < 10^{-4}$). The maximum rank shift is ± 30 positions.

Fixed-income funds are the most upgraded: long-term bonds (TLT) shift +30 positions, high-yield bonds (HYG) +28, and investment-grade bonds (LQD) +24.5. European equities receive the largest downgrades: Italy (EWI) shifts -30 , Spain (EWP) -29 (Figure 1).

Regional corrections are systematic: US-region funds (incl. bonds/REITs) receive moderate corrections (mean $|\Delta| = 8.8$, $n = 23$), European funds the largest (mean $|\Delta| = 14.9$, $n = 9$), followed by emerging markets (9.8, $n = 8$) and Asia-Pacific (7.4, $n = 7$).

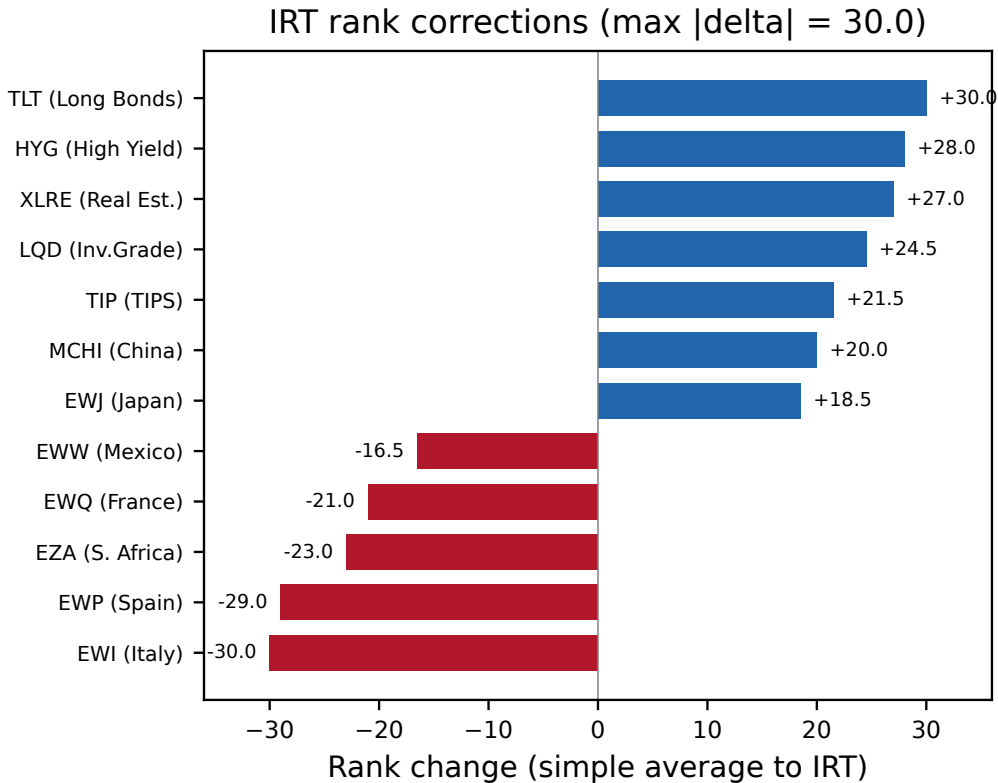


Figure 1: IRT rank corrections for the 12 largest movers. TLT (+30) and HYG (+28) are the most upgraded; EWI (-30) and EWP (-29) the most downgraded.

3.2 Crisis Cross-Validation

Crisis periods have mean $b = -0.710$; normal periods have mean $b = 0.141$. The month-level Mann–Whitney test gives $p = 0.001$. The contiguous-block permutation test gives $p = 0.055$ —marginal at $\alpha = 0.05$, consistent with but not conclusively confirming a difficulty difference after accounting for temporal dependence (Figure 2).

The negative crisis difficulty is consistent with the interpretation that downturns are easier for relative outperformance: when ACWI drops, bonds, gold, and defensive sectors mechanically outperform. The beat rate–difficulty correlation is $r = -0.733$ ($p < 10^{-6}$), providing suggestive construct-validity evidence.

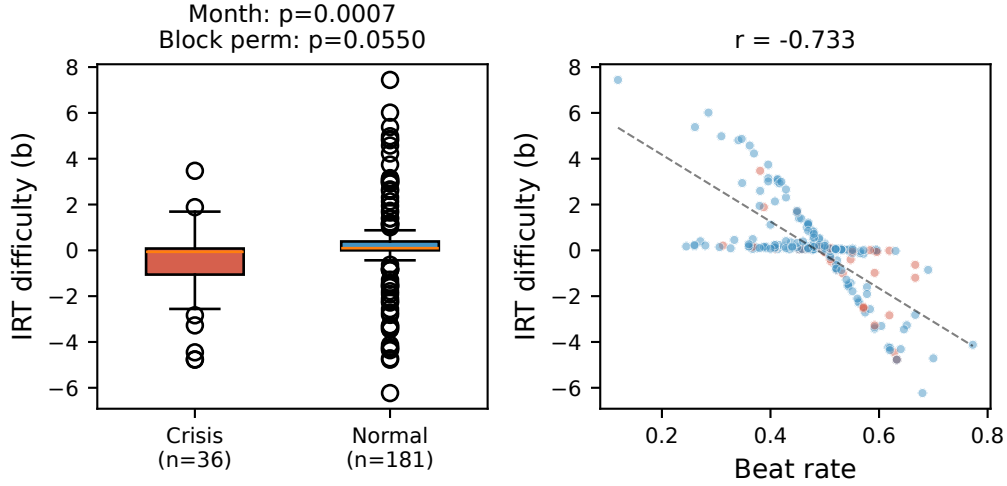


Figure 2: Left: difficulty by crisis status (month $p = 0.001$; block permutation $p = 0.055$). Right: beat rate vs. difficulty ($r = -0.733$). Red = crisis.

3.3 LLTM Decomposition

The LLTM achieves in-sample $R^2 = 0.076$ but cross-validated $R^2 = -0.048 \pm 0.190$, indicating that the 5-feature linear model does not reliably predict out-of-sample monthly difficulty. Feature weights are exploratory: ACWI return ($\eta = +0.25$), credit spread ($\eta = -0.22$), crisis flag ($\eta = -0.18$), and dispersion ($\eta = -0.17$). These should not be treated as confirmatory (Figure 3).

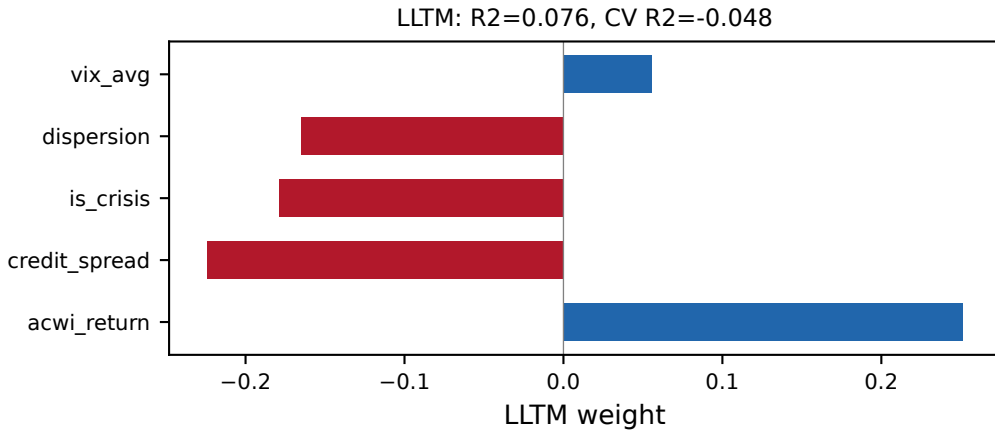


Figure 3: LLTM feature weights (exploratory). Cross-validated $R^2 < 0$.

3.4 PCR Factor Structure

PC1 explains 59.2% of variance, consistent with a dominant common return component, with top-3 PCs at 70.9% (Figure 4). PCR with 3 components predicts difficulty at $R^2 = 0.104$.

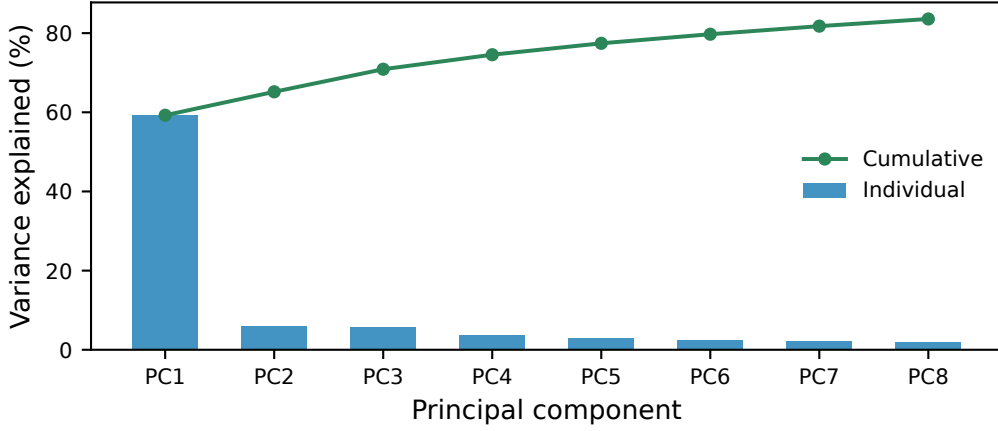


Figure 4: PCA variance explained.

3.5 Sparsity Stress Test

We randomly mask a fraction $s \in \{0, 0.1, \dots, 0.9\}$ of the observed entries, refit a 1PL Rasch model for computational stability, and compare the Spearman ρ of the 1PL and simple-average rankings against the full-data 2PL ranking. This reference ranking is the main IRT estimate, not an external ground truth, so the test measures agreement with the full-data IRT solution; the $s = 0$ level applies no masking and is counted as one of the 10 levels. The canonical run (`experiment.py: 5` masking replicates per level, `L-BFGS-B maxiter 300`) recorded the 1PL refit ahead at 7 of 10 levels (Figure 5); only this count, not the per-level values, is committed as data (`data/canonical_stress.json`). An offline refit (`reproduce_from_derived.py --stress-refit: 2` masking replicates per level, `maxiter 200`; per-level values in `data/stress_refit_levels.csv`) gives 6 of 10: the 1PL refit leads by $+0.003$ to $+0.014$ at $s = 0$ – 0.4 and $s = 0.6$, and trails at $s = 0.5$ (-0.002), 0.7 (-0.002), 0.8 (-0.017), and 0.9 (-0.041). IRT therefore loses at high sparsity in the offline refit. We note that the stress-test refit uses a simpler model than the main 2PL analysis; the 1PL omits discrimination parameters, providing a lower-capacity robustness check rather than an exact replication of the main 2PL estimator.

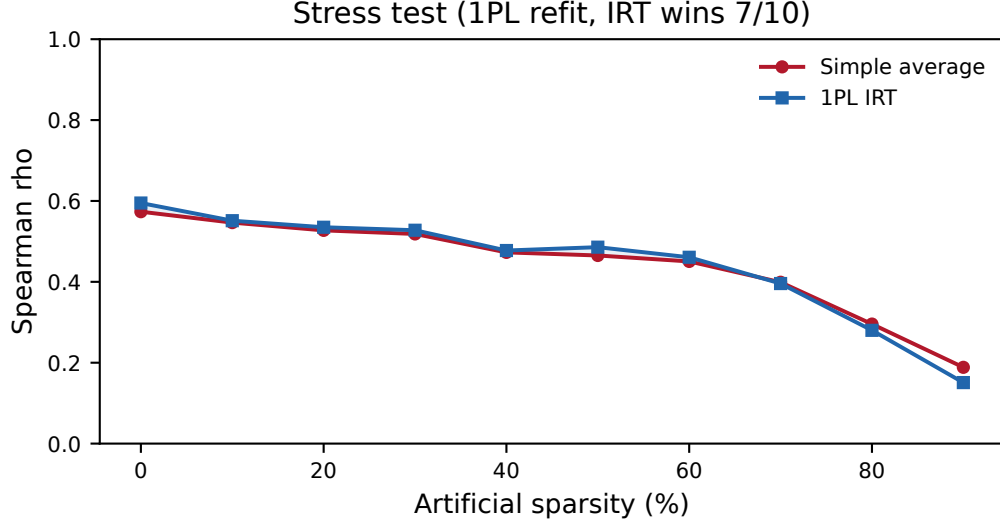


Figure 5: Sparsity stress test using 1PL Rasch refit, canonical run (recorded as 7/10 IRT wins; per-level values not committed as data). Reference ranking: full-data 2PL estimate. The offline refit (6/10; `data/stress_refit_levels.csv`) is reported in the text.

3.6 Regional Analysis

European funds receive the largest corrections (mean $|\Delta| = 14.9$), followed by emerging markets (9.8). Asia-Pacific (7.4) and global/commodity funds (4.7) receive smaller corrections than US-region funds (incl. bonds/REITs; 8.8) (Figure 6).

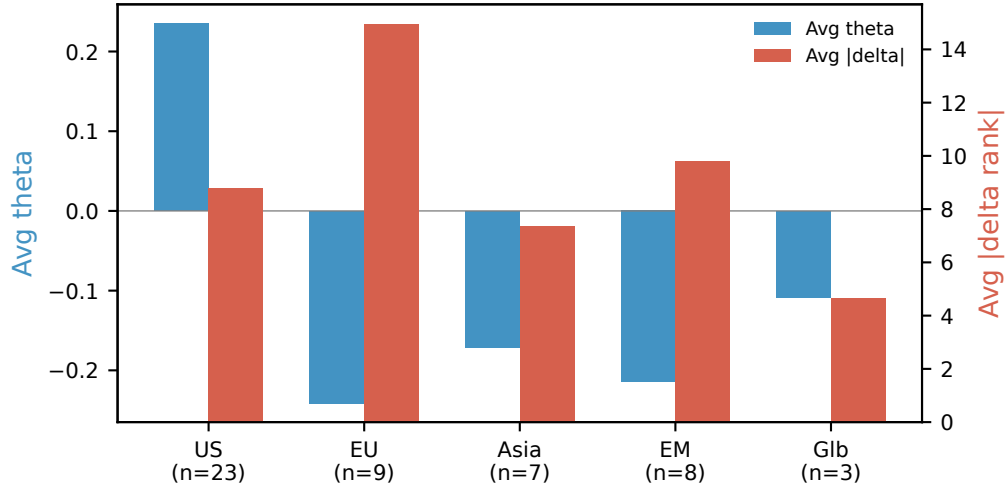


Figure 6: IRT ability and correction magnitude by region.

4 Discussion

Difficulty inversion. Crisis periods appear easier for relative outperformance, consistent with the EFSL pattern where “hard” conditions for observers differ from “hard” conditions for the ranking problem. However, the block-level permutation ($p = 0.055$) indicates this difference is marginal once temporal autocorrelation is accounted for; future work with more crisis episodes or longer history may clarify this.

Comparison to risk-adjusted metrics. Sharpe ratios and factor alphas [Sharpe, 1966, Jensen, 1968, Fama and French, 1993, Carhart, 1997] assess whether funds generate excess risk-adjusted return. IRT assesses whether a fund’s *ranking* is distorted by regime-specific evaluation bias. The two are complementary. Future work should examine whether IRT-corrected rankings predict out-of-sample performance better than factor-based rankings.

Limitations. Unidimensionality and local independence are approximate; formal diagnostics are needed. Because the universe mixes equity, fixed-income, commodity, real-estate, and digital-asset ETFs, estimated period difficulty partly reflects benchmark-relative asset-class exposure rather than a pure market-regime difficulty parameter; crisis “easiness” is partly mechanical when non-equity assets are compared against an equity benchmark. LLTM does not generalize out-of-sample at monthly resolution. The stress test uses a simpler 1PL model. Natural sparsity (5.7%) is low compared to other EFSL domains.

5 Conclusion

IRT rank corrections of ± 30 positions among 50 global ETFs suggest that regime-dependent evaluation effects can materially alter fund rankings. Crisis periods are consistent with being easier for relative outperformance (block permutation $p = 0.055$). In the sparsity stress test (1PL, scored against the full-data IRT ranking), the canonical run recorded IRT ahead at 7/10 levels without committed per-level values, while an offline refit gives 6/10 with margins of at most $+0.014$ and losses at high sparsity (down to -0.041 at 90%). This provides an additional cross-domain application of the EFSL framework.

Code and reproduction artifacts are provided in the accompanying source package.

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